5 Amino 1MQ Available for Research Use Only

(175 -300 mg / day for body recomposition; 25 – 50 mg/day for longevity)

5-Amino-1MQ has demonstrated adipose, metabolic, and aged-muscle effects in specific animal models.

5-Amino-1MQ (5-amino-1-methylquinolinium) is an experimental small-molecule inhibitor of nicotinamide N-methyltransferase (NNMT) with substantial preclinical evidence. [1,2,7–10] Reduced adiposity has been observed in obese-mouse experiments.[2,4] Higher NAD+ was reported in cultured adipocytes and, separately, mouse muscle after NNMTi treatment. Mitochondrial and epigenetic consequences remain context-dependent. [2,3,17]

❓ Unknow safety

⭐ Plausibly effective for body recomposition

⭐ Plausibly effective for longevity

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

This document was updated 12-Oct-26. The most recent version of this document is available at https://pdfs.researcher6076.com/pdfs/5-amino-1mq.pdf

Human pharmacokinetics, efficacy, and safety remain unestablished; chronic consequences of NNMT inhibition are uncertain. [1–5,7–10]

Evidence Summary

The strongest direct evidence concerns selected endpoints in obese or aged mice. The 2021 diet-combination study and 2022 microbiome study are condition-specific; microbiome differences do not establish that a microbiome change mediates benefit. Much of the compound-specific work comes from an overlapping research network, with commercial interests disclosed in several papers. Independent replication and human translation remain important gaps. Broader NNMT findings and newer inhibitor chemotypes should be labeled separately. [4–9,11]

Animal Pharmacokinetics and Human Mechanistic Extrapolations

Animal pharmacokinetic data are available for 5-Amino-1MQ, including oral, intravenous (IV), and subcutaneous (SC) administration. No human PK data have been established. The animal studies show substantial differences between species and routes of administration, particularly for oral bioavailability. [7,10]

Rat PK: Awosemo et al. evaluated IV and oral 5-Amino-1MQ in adult male Sprague-Dawley rats. Oral administration produced a mean Cmax of **2,252 ng/mL**, AUC0–∞ of **14,431 ng·h/mL**, and terminal half-life of **6.9 ± 1.2 hours**. IV administration produced an AUC0–∞ of **3,708 ng·h/mL** and terminal half-life of **3.8 ± 1.1 hours**. Calculated oral bioavailability was **38.4%**. [10] ScienceDirect

Mouse PK: Babula et al. evaluated IV, oral, and SC administration in adult male C57BL/6J mice. A **5 mg/kg IV** dose produced a Cmax of **2,009 ng/mL**, AUC0–∞ of **825 ng·h/mL**, and terminal half-life of **6.3 hours**. A single **25 mg/kg SC** dose produced a Cmax of **7,010 ng/mL**, AUC0–∞ of **7,031 ng·h/mL**, and half-life of **13.3 hours**. After five once-daily 25 mg/kg SC doses, Cmax was **5,130 ng/mL**, AUC0–∞ **4,687 ng·h/mL**, and half-life **12.8 hours**. [7] PubMed Central (PMC)

Oral exposure in the mouse study was much lower. A **30 mg/kg oral** dose produced a Cmax of only **14.5 ng/mL**, AUC0–∞ of **224 ng·h/mL**, terminal half-life of **14.8 hours**, and estimated oral bioavailability of **3.5%**. The investigators considered limited intestinal absorption and substantial first-pass metabolism likely explanations. [7] PubMed Central (PMC)

Thus, reported oral bioavailability ranges from approximately **3.5% in mice to 38.4% in rats**, an approximately elevenfold difference. This large species/study difference makes extrapolation of oral exposure particularly uncertain. The relatively long terminal half-lives also should not be interpreted as equivalent periods of effective NNMT inhibition. In mice, plasma concentrations remained above the reported biochemical mouse NNMT IC50 of 78 ng/mL for only approximately **2–4 hours after parenteral administration**, despite terminal half-lives of 6–13 hours. [7] PubMed Central (PMC)

Human-equivalent-dose comparisons: Standard body-surface-area scaling provides the following approximate HEDs for a **150-lb (68-kg) adult**. These are mathematical comparisons with animal doses, not established human doses. [14]

Animal regimen	Approximate HED for 150 lb
Mouse 5 mg/kg IV PK dose	28 mg
Mouse 10 mg/kg/day	55 mg/day
Mouse 20 mg/kg SC three times daily (60 mg/kg/day)	110 mg per administration; 331 mg/day total
Mouse 25 mg/kg SC PK dose	138 mg
Mouse 32 mg/kg/day	176 mg/day

Body Recomposition Dosing Patterns

The animal literature provides the strongest body-recomposition signal from **higher or more intensive exposure.** The most directly relevant studies used relatively higher exposures in diet-induced-obese mice: the original Neelakantan study used **20 mg/kg SC three times daily for 11**

days (approximately **331 mg/day total HED for a 150-lb human** by BSA scaling) and produced actual weight and adipose-mass loss, while later studies used **32 mg/kg SC once daily** (approximately **176 mg/day HED**) for several weeks and showed reduced adiposity and improved metabolic outcomes; Babula et al. also found effects at **10 mg/kg once daily** (approximately **55 mg/day HED**), with generally greater metabolic effects at 32 mg/kg. [2,4,7]

The plausible SC dose for body recomp goals would be

- **175 – 300 mg daily for several weeks**

Healthspan Dosing Patterns

The strongest healthspan-oriented evidence comes from **lower, sustained once-daily exposure at 10 mg/kg/day for eight weeks**. The best-aligned model is the 2024 aged-mouse study: **10 mg/kg SC once daily for eight weeks** (approximately **55 mg/day HED**), with treatment spanning ages 22–24 months and improving grip strength and other muscle-related outcomes, particularly when combined with exercise. [8] The 2019 aged-muscle injury study provides additional support but used a different pattern, **5 or 10 mg/kg SC twice daily** around experimentally induced muscle injury, making it more directly relevant to regeneration than general healthspan. [9]

However, The evidence creates a plausible concern that these substantially smaller doses could be ineffective because they may:

- produce a lower peak;
- spend little or no time near the effective concentration range;
- penetrate some target tissues inadequately;
- cause only weak, short-lived reductions in 1-MNA.

The mouse efficacy study provides some support for a minimum threshold dose: **10 mg/kg/day SC (HED 55 mg) produced much less metabolic efficacy than 32 mg/kg/day (HED 176 mg)**, with the lower dose failing to prevent body-weight and fat-mass gain. Some biochemical outcomes nevertheless changed at the lower dose, showing that different endpoints and tissues can have different exposure requirements, so **smaller doses could still plausibly work if partial, intermittent NNMT inhibition is sufficient for a desired healthspan endpoint**.

A plausible healthspan dose is

- **25 - 50 mg daily**

A healthspan regime tends to be continuous, but, importantly, no study tested lifelong treatment, very-low-dose continuous exposure, intermittent longevity cycles, or whether maintaining NNMT inhibition indefinitely is preferable to periodic inhibition.

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Effective animal treatment did not require demonstrated continuous or complete NNMT suppression. Both 5 and 10 mg/kg regimens produced biological effects in aged mice, and 10 mg/kg once daily remained effective despite PK evidence indicating that pharmacologically substantial NNMT inhibition was unlikely to persist throughout the 24-hour dosing interval. Cellular studies likewise show meaningful biological effects with partial suppression of the NNMT product 1-MNA. These findings **support the plausibility that intermittent or partial NNMT inhibition may be sufficient for healthspan-related effects**.

Continuous NNMT Inhibition and Long-term Safety

No study has established the safety or benefit of maintaining pharmacologic NNMT inhibition indefinitely. The available 5-Amino-1MQ studies nevertheless provide **some reassurance for sustained exposure over weeks**, including eight weeks of once-daily treatment in very old mice without an obvious toxicity phenotype. [8] Importantly, **successful animal regimens did not necessarily maintain plasma concentrations throughout the dosing interval**, suggesting that continuous 24-hour enzyme inhibition may not be required for metabolic or muscle benefits.

Mechanistically, chronic *partial or intermittent* inhibition may be more defensible than attempting continuous maximal suppression. NNMT competes with NAD+ salvage for nicotinamide and consumes SAM, so reducing excessive NNMT activity could favor NAD+ availability and methyl-donor conservation. However, NNMT also produces 1-MNA, which has demonstrated protective antioxidant, anti-inflammatory, vascular, and tissue-specific effects, and recent studies show that NNMT activity can be beneficial under some physiological and stress conditions. [12,13] Consequently, **complete lifelong suppression could theoretically sacrifice beneficial NNMT/1-MNA functions or produce cumulative effects on nicotinamide and methylation metabolism**. The **existing evidence supports sustained moderate NNMT inhibition for healthspan more strongly than it supports continuous maximal inhibition**.

Whether lower daily exposure, intermittent inhibition, or treatment cycles provide the best balance of benefit and safety remains unknown.

Plausible Benefits

- **Adiposity and fat mass:** compound-specific mouse studies support an effect on adipose biology. [2,4]
- **NAD+ and energy metabolism:** cell experiments and NNMT knockdown studies support links to NAD+ salvage, SAM availability, and energy metabolism. Genetic target manipulation is not equivalent to administering 5-Amino-1MQ. [2,3]
- **Glucose regulation:** improved glucose handling and reduced hyperinsulinemia were reported in diet-induced obese mice. [7]
- **Muscle regeneration and function:** aged-mouse studies reported improved recovery after injury and selected functional outcomes, including grip strength. [8,9]

- **Obesity-associated hepatic steatosis:** improvements have been reported in mouse disease models, including in combination with dietary change. [4,7]

Potential Applications

- **Obesity and adiposity reduction:** investigational indication supported by mouse experiments. [2,4]
- **Metabolic dysfunction and age-related muscle impairment:** research indications; human efficacy unestablished. [7–9]

Cautions

- **Pregnancy, conception, and breastfeeding:** safety unknown. A September 2026 NNMT-knockdown study found impaired endometrial decidualization and fewer mouse implantation sites; this is target-level evidence, not reproductive toxicology of 5-Amino-1MQ. [13]
- **Hepatic and renal impairment:** human clearance and dose adjustments are unknown.
- **Drug interactions:** no established human interaction profile. Altered nicotinamide and methyl-donor metabolism raises theoretical concerns. [7]
- **Insomnia or stimulation:** anecdotal reports.

Reported or Possible Side Effects

- **Gastrointestinal symptoms:** mentioned in informal reports.
- **Sleep disturbance:** informal reports.
- **Restlessness, warmth, or sweating:** unverified anecdotes.
- **Altered methyl-donor metabolites:** a mechanistic possibility rather than a demonstrated human adverse effect. Direction and clinical significance may differ across tissues. [3,13]

Drivers of Diminished Response and Mitigation Strategies

Whether responses diminish with sustained treatment is unknown. As a small-molecule enzyme inhibitor, 5-Amino-1MQ does not have the same expected desensitization or immunogenicity mechanisms as receptor agonists or biologics. Changes in NNMT expression, substrate supply, clearance, or downstream pathways remain possible but unproven explanations.

Known Unknown Safety Risks

No published human administration study establishing human PK, efficacy, or controlled safety exist.

- **Hepatic effects are context-dependent.** NNMT participates in liver metabolism, but inhibition does not itself prove hepatotoxicity. In a 2025 ischemia–reperfusion study, NNMT/1-MNA signaling was protective; this target-level result cautions against assuming inhibition benefits every liver condition. [12]

- **One-carbon and methylation balance:** inhibiting a SAM-consuming enzyme can alter SAM/SAH flux and downstream methylation. The 2026 decidualization work provides a concrete example of potentially unfavorable consequences of NNMT deficiency in a specific tissue. [3,13]
- **NAD+ and nicotinamide biology:** more available nicotinamide need not yield more NAD+ in every tissue. Salvage capacity, substrate concentrations, and NAD+ consumption matter; nicotinamide can also inhibit NAD+-dependent enzymes under some conditions. A favorable whole-body longevity effect cannot be inferred from pathway direction alone.
- **Off-target and long-term risks:** a 10 μM screening assay reported 67.4% MAO-A inhibition. [7]

Healthspan Benefits and Rationale

The 2019 injury study and 2024 aged-muscle study measured regeneration and function. In the latter, mice aged 22 to 24 months received 10 mg/kg/day subcutaneously for eight weeks; grip strength improved and treatment plus exercise produced additional benefit. Effects were endpoint-specific and do not demonstrate reversal of aging. [8,9]

Preclinical justification: Preserving muscle function and reducing pathological adiposity could plausibly improve healthspan. But benefits in obese or injured animals may not apply to healthy, lean adults.

Chanvillard et al. (January 2026) linked NNMT to tubular senescence and fibrosis in human kidney specimens and tested 5-amino-1-methylquinolinium in cultured tubular cells and human kidney organoids, supporting a kidney-specific hypothesis, not generalized senolysis or healthspan extension. [15]

Dong et al. (2025) used NNMTi (HY-131042, a supplier-identified 5-amino-1-methylquinolinium iodide reagent) at 10 mg/kg/day intraperitoneally in a hindlimb-ischemia model. Strength, power, and work improved without improved perfusion or capillary density. Human muscle analyses were observational, not drug treatment. Liang et al. (2024) separately reported improved grip strength and muscle-related measures in naturally aged and D-galactose-treated mice, with increased muscle NAD+ and AMPK/PGC1α markers. [16,17]

Mechanistically, NAD+ salvage and methyl-donor conservation offer hypotheses for metabolic resilience and cellular repair without establishing that persistent NNMT suppression improves aging. Findings involving genetic knockdown, fibrosis, tumor biology, or different inhibitors suggest 5-Amino-1M has plausible efficacy for healthspan. [3,11]

However, NNMT and 1-MNA can have beneficial functions in particular tissues or stress states. Recent liver and reproductive findings argue against treating NNMT as uniformly harmful. A

healthspan claim, even just plausible healthspan improvement claims, requires consideration of NNMT benefits and these tradeoffs. [12,13]

Biological mechanism

NNMT transfers a methyl group from S-adenosylmethionine (SAM) to nicotinamide (NAM), producing 1-methylnicotinamide (1-MNA) and S-adenosylhomocysteine (SAH). Inhibition can preserve NAM for NAD+ salvage and change methyl-donor flux. 5-Amino-1MQ inhibits NNMT in biochemical and cellular assays, with low-micromolar potency in early work. Connections to sirtuins, polyamine flux, epigenetic state, and energy metabolism are supported to varying degrees by cell, genetic, and pharmacological studies; they are not a fully demonstrated causal chain in humans. [1–3]

Conclusions

5-Amino-1MQ is a preclinical research compound with a rationale for studying metabolic disease and age-related muscle dysfunction. Neither human body recomposition nor longevity efficacy is more than plausible. Animal HED calculations do not establish human dosing, and smaller continuous exposure remains an untested PK/PD and safety hypothesis.

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